Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/06/2024
PI Name:	Laurie, Steven Ph.D.		
Project Title:	Measurements of Transcutaneous PCO2 with TCM5 Monitor During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) (3) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	steven.laurie@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-244-0029
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Cardiovascular and Vision Laboratory		
PI Address 2:	2400 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2022	**End Date:**	05/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/31/2025 per C. Ribeiro/HHC (Ed., 10/24/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ryder, Valerie Ph.D. (NASA Johnson Space Center) Lytle, Jason Ph.D. (KBR/NASA Johnson Space Center) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Flynn-Evans, Erin Ph.D. (NASA Ames Research Center) Macias, Brandon (NASA Johnson Space Center) Carter, Katrina (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	While the partial pressure of carbon dioxide (PCO2) on the International Space Station (ISS) remains below levels expected to cause negative health impacts terrestrially, it is elevated above typical terrestrial levels and some crewmembers report symptoms that resolve or improve when ambient PCO2 levels are reduced. Air quality data for volatile organic compounds indicate good ventilation and mixing across the ISS stack; however, it is possible that crewmembers are exposed to elevated inspired PCO2 levels during certain activities, such as extended operations in the microgravity science glovebox, exercise, or during sleep in crew quarters which are not reflected by the cabin average concentration. This has the potential to increase arterial PCO2 levels and contribute to various negative health outcomes. However, measures of arterial PCO2 throughout the day and night have not been collected in crewmembers on the ISS. The overall objective of this study is to characterize the arterial PCO2 in crew conducting normal daily activities during spaceflight and to determine if variability in CO2 levels contribute to other negative outcomes. To do this, we will test the following specific aims: (1) characterize the range of transcutaneous partial pressure CO2 (PTCCO2) levels in crewmembers conducting normal daily activities on the ISS; (2) determine if nightly PTCCO2 levels are elevated to a greater degree than occurs on Earth, and if these are associated with disrupted breathing during sleep; and (3) determine if there are associations between daily or nightly PTCCO2 levels and Spaceflight Associated Neuro-ocular Syndrome (SANS) findings, CO2-like symptoms, or changes in mood or cognitive performance.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. In May of 2020, the NASA Human Research Program (HRP) asked the Principal Investigators (PIs) of this proposal to lead a Technical Interchange Meeting (TIM) with subject matter experts external to NASA to discuss gaps in knowledge related to CO2 exposure in crewmembers during spaceflight. The primary recommendation that emerged from this meeting was the need to better monitor arterial PCO2 levels in crewmembers during spaceflight and to relate PCO2 measures to physiologic or cognitive outcomes to determine if the mildly elevated inspired levels of CO2 are causing negative impacts to crewmembers. The Human Health and Countermeasures (HHC) Element requested the PIs to develop this proposal to address the recommendation from the TIM. The direct assessment of arterial CO2 levels in crew during spaceflight represents a completely unexplored factor that may help explain the variability in magnitude and/or individual susceptibility in the development of SANS and other neurologic findings. This project will provide highly valuable data to the HRP and Office of the Chief Health and Medical Officer, and recommendations for evaluating NASA's CO2 Standard.
Research Impact/Earth Benefits:	This research study will utilize a non-invasive CO2 monitor to evaluate the CO2 levels in astronauts throughout the day and night while living on the International Space Station. These data will provide insight into the effects of mildly elevated CO2 levels on the physiology of healthy individuals.
Task Progress:	May 2025 Update: Project has closed. Per NASA Johnson Space Center (JSC) no final performance report or data deliverables forthcoming (Ed., 5/28/25). ---------- • The NASA Institutional Review Board (IRB) approved this study protocol in December 2023. • NASA Research Operations and Integration (ROI) developed a Feasibility Assessment (FA) that was presented to the Human Research Program in July 2024. The PI team is waiting on the outcome of the FA before initiating implementation activities.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Abstracts for Journals and Proceedings	Carter KJ, Betts A, Ryder VE, Möller F, Lovering AT, Flynn-Evans E, Macias BR, Laurie SS. "Measurements of transcutaneous PCO2 with TCM5 monitor during spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024